

Debian Trixie on a 2017 iMac 5K: Drivers & Configuration

Display & Graphics: Retina 5K and AMD Radeon Pro

Retina 5K Resolution & Scaling: The iMac's 27" 5K panel (5120×2880) can be tricky on Linux. By default, you may only see lower resolutions (like 2560×1440 or 3840×2160) because the internal display is actually treated as a dual-tiled monitor in hardware. Even Windows in Boot Camp doesn't easily achieve full 5K without Apple's drivers ¹ ². Linux support for the tiled 5K is limited – the AMD GPU driver sometimes fails to drive the native 5K mode reliably (screen corruption or blank output can occur) ³. A common workaround is to run at **2560×1440** with **2× HiDPI scaling**, which effectively gives a "Retina" experience with sharp text. In the MATE desktop, enable **HiDPI mode** (e.g. via MATE Tweak or Appearance settings) to scale the UI by 2× ⁴. This makes the desktop comfortable to use on the high-DPI screen. Fractional scaling (e.g. 1.5×) isn't directly available in MATE's settings (only 100% or 200%), so 200% is recommended for a true Retina look. If you attempt to force 5120×2880 and encounter problems, you can explicitly limit the resolution to 2560×1440@60Hz in GRUB's kernel parameters (e.g. `video=2560x1440@60`) to avoid the GPU driving the unstable 5K mode ³.

AMD Radeon GPU (Radeon Pro 570/575/580): These Polaris-family GPUs are well-supported by the open-source **AMDGPU** driver included in the kernel ⁵. To get full 2D/3D acceleration and proper high-resolution output, you **must install the proprietary firmware blobs** for AMD cards. On Debian, enable the "non-free-firmware" component in apt sources and install the package `firmware-amd-graphics` ⁶. Without this firmware, the GPU may boot in VESA mode or suffer **poor performance** ⁷. Also ensure Mesa drivers and X11 drivers are installed (Debian's `xserver-xorg-video-amdgpu`, `libgl1-mesa-dri`, etc., but these are usually present by default) ⁸ ⁹. The open driver is sufficient for gaming, compositing, and general use. Debian Trixie's kernel and Mesa will support the Radeon Pro 5xx series out of the box – no need for AMD's legacy fglrx or even AMDGPU-Pro unless you have a specific use-case. If you ever require AMD's proprietary stack (AMDGPU-Pro OpenGL/Vulkan), it can be installed manually, but for most users the open stack is recommended.

External Displays: If you connect external monitors via Thunderbolt/DisplayPort, the AMDGPU driver should handle them. Keep in mind the 2017 iMac's GPU can drive additional displays, but driving *another* 5K monitor might be limited by bandwidth. Standard 4K monitors should work fine with appropriate scaling.

Wi-Fi (Broadcom) & Bluetooth

Broadcom Wi-Fi: The 2017 iMac uses a Broadcom wireless chipset (Apple's "AirPort" card, likely a BCM4360-series or BCM43602 802.11ac device). Out of the box, Debian won't have a driver for this because Broadcom chips often **require proprietary driver or firmware**. There are two driver options:

- **Broadcom-STA (wl) Driver:** This is Broadcom's proprietary Linux driver, which supports many Apple Broadcom chipsets (including BCM4360) ¹⁰ ¹¹. On Debian, it's provided by the package `broadcom-sta-dkms` (in the non-free section). After enabling non-free, install it with apt; this DKMS module builds the `wl.ko` driver and blacklists conflicting open-source drivers ¹² ¹³. For example:

```
sudo apt update
sudo apt install broadcom-sta-dkms linux-headers-$(uname -r)
```

This will compile and insert the Broadcom `wl` module (you may need to reboot or manually `modprobe wl` after installation) ¹⁴ ¹⁵. The STA driver should make the iMac's Wi-Fi interface (e.g. `wlan0`) functional. Be aware of known issues – for instance, the Broadcom STA driver can have trouble with WPA3 networks ¹⁶ (fallback to WPA2 if needed).

- **Open-Source `brcmfmac` Driver:** Newer Broadcom chips (like BCM43602) are supported by the kernel's `brcmfmac` module *if* the correct firmware is provided. Debian's `firmware-brcm80211` package supplies these firmware blobs ¹⁷. In many cases, installing `firmware-brcm80211` and rebooting will allow the `brcmfmac` driver to handle the Wi-Fi. The open driver is preferred by some distributions (for example, the T2 Mac/Linux community explicitly recommends *against* using `wl` on newer Macs in favor of `brcmfmac` with firmware ¹⁸). If you go this route, ensure any Broadcom STA/`wl` driver is removed or blacklisted to avoid conflicts.

In summary, **enable Debian's non-free-firmware** and install the Broadcom Wi-Fi firmware package at minimum. If the Wi-Fi still isn't working, use the STA driver. Once the correct driver+firmware is in place, the wireless should work normally (you can then configure SSIDs as usual). *Tip:* If you have no network after install, connect via Ethernet temporarily – Debian's `update-manager` or `Additional Drivers` tool (on Ubuntu/Mint) can detect the Broadcom and offer to install the driver automatically over Ethernet ¹⁹.

Bluetooth: The iMac's Bluetooth is part of the same Broadcom module (combo Wi-Fi/Bluetooth card). It is supported via the kernel's `btusb` driver. Like the Wi-Fi, it likely needs a Broadcom firmware file to be loaded for full functionality (these `.hcd` firmware files are also included in Debian's `firmware-brcm80211`). After installing the firmware package, the Bluetooth adapter should be recognized by BlueZ. In practice, Bluetooth on iMac is usually **plug-and-play once firmware is present** – you can check `dmesg | grep -i bluetooth` to see if it loaded firmware. No proprietary BT driver is needed.

Special notes: On first boot, you'll need a USB keyboard/mouse to set things up if your Apple Magic Keyboard/Mouse aren't paired yet. Once the Bluetooth is up, you can pair your Apple peripherals. Also, **Bluetooth audio** (A2DP headsets, etc.) will work via BlueZ/PulseAudio (or PipeWire) as on any Linux PC.

Audio: Internal Speakers, Headphone Jack & Mic (Cirrus Logic CS8409)

Audio support on this iMac requires special attention. The **Cirrus Logic CS8409** audio codec (with Apple's custom amplifiers) is *not* fully supported by the stock kernel's HDA driver. Out of the box, you'll likely see **"Dummy Output"** with no sound from speakers or headphone jack. This is a known issue on 2016–2019 era Macs. The Linux kernel did introduce a basic codec driver (`snd_hda_codec_cs8409`), but it doesn't initialize the iMac's audio properly (hence no output) ²⁰. To get audio working, you'll need to install a **community-developed driver/patch** for the Cirrus 8409 codec:

- One popular solution is the open-source project `snd_hda_macbookpro` by developer Davidjo (and variants by egorenar). This is an out-of-tree kernel module specifically for Macs with the CS8409 + Apple amp setup ²¹ ²². It programs the codec and amps correctly for sound output. Users have reported success compiling and using this on iMacs ²³ ²⁴. There is even a Debian/Ubuntu DKMS package in circulation (e.g. on Linux Mint forums) to make installation easier.

How to install the audio driver: The general steps are: download the source (for example, clone the GitHub repo `egorenar/snd-hda-codec-cs8409` or `davidjo/snd_hda_macbookpro`), compile it against your current kernel, and install the module. For instance, one can do:

```
git clone https://github.com/egorenar/snd-hda-codec-cs8409.git
cd snd-hda-codec-cs8409
make
sudo make install
reboot
```

After reboot, the new codec driver should take over. Indeed, users confirm that after installing this driver, **analog sound output works** on the iMac27 (sound comes from internal speakers and headphone jack) ²³ ²⁴. Debian Trixie's kernel might include an incomplete CS8409 driver, so adding this ensures full support.

Internal Microphone: Unfortunately, **microphone support remains problematic**. The CS8409 driver from the community is still output-focused – capturing from the internal mic is either very quiet or not wired up fully ²⁵ ²². In practice, many users find the iMac's built-in mic *does not work* reliably on Linux ²⁶. You may need to use an external microphone or a Bluetooth headset if you require input. (Some have had minor success by boosting the capture volume, but the quality is low.)

Volume Keys and Outputs: Once sound is working, you'll use the standard ALSA/PulseAudio controls. The iMac has a 4-speaker system (left/right tweeters and woofers); the custom driver presents this as a normal stereo output but actually drives all speakers ²⁷. Use the **"Analog Stereo" output profile** in PulseAudio ²⁸. The headphone jack should automute speakers when plugged in. If you have the Apple USB-C to 3.5mm adapter or other devices, those use standard USB audio drivers. The volume/media keys on an Apple keyboard should adjust audio via GNOME/MATE as usual (if not, you can map them).

Note: Keep the custom audio module source handy – if you update your kernel, you may need to recompile or reinstall the module (DKMS can automate this). There is hope that upstream Linux will merge full CS8409 support, but until then this is the way to get sound.

Webcam (FaceTime HD Camera)

The 2017 iMac's built-in webcam (often labeled "FaceTime HD") is **not plug-and-play** with Linux – it uses a proprietary Broadcom controller (internal PCIe) that doesn't have a native kernel driver ²⁹. Out of the box, your iMac's camera will **not be detected** (e.g. `ls /dev/video*` won't show it). But don't worry: the open-source community has provided a solution in the form of the **facetimehd** driver. This is a reverse-engineered driver for the Broadcom 1570 chipset found in Macs ³⁰ ³¹.

Installation of Webcam Driver: Debian might have a DKMS package (`facetimehd-dkms`) in non-free, but if not, you can build it manually. The process involves two parts:

1. **Firmware:** First, obtain the firmware for the camera. There's a utility on GitHub (patjak's `facetimehd-firmware` repo) that extracts the firmware blob from Apple's BootCamp driver. Running `make && sudo make install` in that repo will install a file (e.g. `/lib/firmware/facetimehd.bin`). This firmware is required for the device to operate.
2. **Kernel Module:** Next, compile the driver module. Clone the `bcwc_pcie` project by Patjak (or Xorx), then run `make && sudo make install`. This builds the `facetimehd.ko` module and installs it to your kernel modules. Load it with `sudo modprobe facetimehd`.

After these steps, the FaceTime HD camera should be recognized by the system ³² ³³. You can verify with `dmesg | grep facetimehd` (it should indicate firmware loaded and camera ready) ³⁰. Testing is easy: for example, install `mpv` or use `ffplay` to open `/dev/video0`, or use Cheese/Kamoso – you should see the webcam feed ³⁴.

Keep in mind this driver is **out-of-tree** (not in the official kernel), so future kernel updates might require re-building it. It's a good idea to script the installation or use DKMS if available. Once set up, however, the iMac's webcam works well for video calls, etc., just like on macOS.

(Side note: "iSight" vs FaceTime HD – older pre-2010 Macs had a USB iSight camera that was supported in kernel; the iMac 2017's camera is the newer Broadcom which needs this special driver. So, this is a known hurdle on modern Intel Macs ²⁹.)

Power Management (Sleep, Fans, Brightness, etc.)

Sleep/Suspend: Basic ACPI suspend **does** work on the iMac, but there are some caveats. Since this model lacks the Apple T2 chip, it uses standard ACPI S3 sleep (suspend-to-RAM). You can suspend the system via the desktop or `systemctl suspend`, and it should enter a low-power state and blink the power LED as usual. **However, a known issue** on some iMacs is the so-called "sleep of death" or spontaneous reboot on resume. For example, users have observed that after one successful sleep, the *next* attempt causes a reboot instead of a proper wake ³⁵. This appears to be a bug triggered by certain drivers (possibly Bluetooth or GPU re-init). If you encounter this, one workaround is to disable problematic devices before suspend (for

instance, some have had to unload the Broadcom wifi or USB controllers). In general, test suspend/resume thoroughly: if it's unstable, you might opt to disable automatic sleep and just shut down when not in use, or try kernel parameters (e.g. `mem_sleep_default=deep` which ensures S3 is used). Check community forums for "iMac suspend reboot" for any emerging fixes – it's a known quirk ³⁵ .

Hibernate: The iMac has plenty of RAM and a fast SSD; hibernation (suspend-to-disk) is less commonly used and would require swap space setup. It's typically not configured by default on Debian. If suspend is unreliable, you could configure hibernate, but on desktops it's usually unnecessary.

Fan & Thermal Management: The iMac's cooling is controlled by Apple's SMC (System Management Controller). By default, the firmware sets fan speeds automatically based on temperature sensors, even under Linux. The kernel's `applesmc` module will load to expose those sensors (you can see temps/fan RPM via `sensors` command once you install `lm-sensors`). Usually, you **don't need manual fan control** – the system will ramp up fans to protect itself. Some users, however, prefer to fine-tune fan behavior (to run cooler or quieter). Utilities like `macfanctld` or `mbpfan` can be used for this. These daemons read Apple's sensors and adjust fan RPM via the applesmc interface ³⁶ . They're not strictly necessary, but are available if you want custom fan curves. In most cases, the default SMC control keeps the iMac adequately cooled (just note that Linux might not aggressively boost fans, potentially running warmer but still within safe range; a fan daemon can make it more "aggressive" if desired ³⁷).

Power Consumption: On a desktop, this is less of a concern than on a laptop. The iMac will not significantly downclock or save power beyond what the CPU and GPU drivers naturally do. Still, you have full support for CPU frequency scaling (Intel P-state driver), and the Radeon GPU will use DPM (Dynamic Power Management) to clock down when idle. Tools like `tlp` are mainly for laptops and aren't needed here.

Backlight Brightness: Adjusting the screen brightness on an all-in-one desktop isn't as straightforward as on laptops. The iMac's LCD backlight may be controlled via DDC or a custom Apple mechanism. The usual laptop brightness keys (F1/F2 on an Apple keyboard) might not work by default. You can adjust brightness in a few ways: - Via software: using `xrandr --brightness` (which is just gamma scaling – not true backlight control, but an option). - Via DDC: some have had success using a tool like `ddcutil` to send DDC commands to adjust the backlight on external monitors; it *might* work on the iMac's internal display if exposed. - There is an unofficial utility `displaybrightness` that interfaces with Macs' brightness. Also, certain desktop environments (KDE, etc.) might have support through the `light` utility or ACPI.

Make sure the kernel module `apple_bl` is loaded – it sometimes provides an interface for Apple backlight. Check if `/sys/class/backlight/apple_backlight/` exists; if so, you can echo values to `brightness` . If not, you may simply rely on the panel's auto brightness (though iMacs don't have ambient light sensors like MacBooks do). In short, brightness control may require a bit of tinkering, but since iMac screens are usually set at a comfortable level or adjustable via GUI sliders, it's manageable.

Other Devices: The iMac's USB ports, Thunderbolt 3 ports, SD card reader, and ethernet should all work with standard drivers. Ethernet is Broadcom as well, but that one is usually supported by the open `tg3` or `bnx2x` driver (just ensure you installed `firmware-misc-nonfree` if needed for Ethernet firmware). The SD card reader in iMacs is typically USB-based (Realtek), which works with the `rtssx` driver.

EFI and Booting: Since you're doing a dedicated Linux install, note that the iMac uses UEFI. Debian Trixie should install in UEFI mode without issues. The GRUB bootloader will reside in the EFI partition. One tip: keep the Apple EFI firmware up-to-date (if you had macOS, it would update through OS upgrades – if not, consider temporarily reinstalling macOS to get firmware updates or use Apple's Firmware Update packages). Modern Apple firmware is generally fine with Linux, but updated firmware can improve things like Thunderbolt compatibility or fix boot quirks.

Proprietary Drivers vs Open Source – Summary

To summarize the driver situation for this iMac:

- **AMD GPU:** Use open-source *amdgpu* driver + **non-free firmware** package ⁶. (No need for a proprietary driver; AMD's open driver is excellent and fully accelerated on this hardware.)
- **Wi-Fi:** Likely needs a **proprietary Broadcom** driver or at least proprietary firmware. Install either `broadcom-sta-dkms` (wl driver) ¹⁴ or the `firmware-brcm80211` package to use the open `brcmfmac` driver ¹⁷. This is the one area where you almost certainly need *non-free software* for full functionality.
- **Bluetooth:** Uses the kernel's open driver (btusb) but **requires firmware** (provided by the same Broadcom firmware package). No separate proprietary driver needed.
- **Audio: Community "out-of-tree" driver** required (since the open driver in kernel is incomplete). This is open-source but not included by default. You'll compile and install it manually (or via DKMS from a community repository) ³⁸ ³⁹. No official proprietary driver exists for the Cirrus codec; the solution comes from reverse-engineered open code.
- **Camera: Open-source driver (facetimehd)** required, plus proprietary firmware extracted from Apple's files ³¹ ⁴⁰. All user-space, as Apple doesn't provide a Linux driver. After installation, it's fully functional.
- **Others (USB, etc.):** All use open-source kernel drivers (no action needed).

Finally, **check community resources** specific to Linux on Mac. The Arch Linux Wiki's Apple section and the *t2linux* project (though focused on T2 Macs) have useful tips for iMac19,1 which are applicable to iMac18,3 (2017 model) as well. Additionally, threads on forums (Ubuntu, Mint, Arch) about iMac 5K installs contain troubleshooting gems for this hardware. For example, users have shared patched kernel modules and config tweaks (like the audio driver and suspend quirks) ²³ ³⁵. Leveraging these community solutions will save you time.

Known Quirks Recap:

- Expect to spend some time getting **audio** working (compiling the driver). Internal mic might remain an issue ²⁶.
- The **first boot** post-install might have no Wi-Fi – prepare by having Ethernet or the driver packages ready (as noted, Debian can load Broadcom drivers if it detects them and you allow non-free during install).
- The **Retina display** will work, but likely at a scaled resolution. Full 5K tiled mode is an edge-case not fully solved in Linux – but 2560×1440 HiDPI is gorgeous and practical.
- **Suspend** may not be 100% reliable; test it, and consider disabling auto-suspend if it misbehaves to avoid data loss on resume issues ³⁵.

- **Bluetooth** should be fine for peripherals; if you have issues pairing the Apple Magic Mouse/Keyboard, consider using a USB connection or re-pair in macOS first (they can remember pairing for use in Linux in some cases).

With the above drivers, firmware, and tweaks in place, Debian Trixie should run **smoothly on the 2017 iMac 5K**, delivering full functionality. Many users have successfully turned these iMacs into powerful Linux workstations ⁴¹ ⁵, so you'll be in good company. Good luck, and enjoy your high-resolution Linux desktop on Apple hardware!

Sources & References:

- Debian Wiki – AMD Graphics Driver Setup ⁶ ⁷ (firmware requirement for amdgpu)
- ArchWiki – Notes on iMac 5K resolution issues ³ (suggesting 1440p as workaround)
- Ubuntu MATE Community – HiDPI Scaling in MATE ⁴
- Debian Wiki – Broadcom Wi-Fi (wl driver) ¹⁴ ¹³; brcmfmac firmware ¹⁷
- Reddit r/pop_os – User tips for Linux on Mac (mentions Wi-Fi, BT, mic issues) ²⁶
- LinuxQuestions forum – Solving iMac (Cirrus CS8409) audio with custom driver ³⁸ ³⁹
- Arch Linux Forum – iMac17,1 audio fix using egorenar's CS8409 driver ²³ ²⁴
- GitHub (davidjo's repo) – Notes on Cirrus driver capabilities ⁴² ²²
- Andrea Fortuna's Blog – FaceTimeHD webcam on Debian (explains driver & firmware) ³¹ ⁴⁰
- AskUbuntu – iMac suspend issue description ³⁵ (suspend causing reboot)
- The Linux Shop Blog – Linux on iMac 2017 (hardware overview and challenges) ⁵ ⁴³

¹ ² running ubuntu on the 5k imac - Apple Community

<https://discussions.apple.com/thread/6626315>

³ Laptop/Apple - ArchWiki

<https://wiki.archlinux.org/title/Laptop/Apple>

⁴ Screen scaling on high resolution display : r/UbuntuMATE - Reddit

https://www.reddit.com/r/UbuntuMATE/comments/chacsk/screen_scaling_on_high_resolution_display/

⁵ ⁴¹ ⁴³ Linux Operating System for Your iMac Mid 2017 | Linux DVD Shop - UK Linux Shop supplying Linux on CD, DVD and USB delivering to the UK and worldwide with 24hr dispatch.

<https://thelinuxshop.co.uk/linux-operating-system-for-your-imac-mid-2017-p-825.html>

⁶ ⁷ ⁸ ⁹ AtiHowTo - Debian Wiki

<https://wiki.debian.org/AtiHowTo>

¹⁰ ¹¹ ¹² ¹³ ¹⁴ ¹⁵ ¹⁶ wl - Debian Wiki

<https://wiki.debian.org/wl>

¹⁷ brcm80211 - Debian Wiki

<https://wiki.debian.org/brcm80211>

¹⁸ Wi-Fi and Bluetooth - t2linux wiki

<https://wiki.t2linux.org/guides/wifi-bluetooth/>

¹⁹ ²⁶ Completely Replacing MacOS with Pop! OS on an 2017 iMac? : r/pop_os

https://www.reddit.com/r/pop_os/comments/17ncox0/completely_replacing_macos_with_pop_os_on_an_2017/

20 Audio Card Cirrus Logic CS8409 on Ubuntu 21.10 iMac19,2 not ...

<https://askubuntu.com/questions/1369422/audio-card-cirrus-logic-cs8409-on-ubuntu-21-10-imac19-2-not-recognized>

21 22 25 27 28 42 GitHub - davidjo/snd_hda_macbookpro: Kernel audio driver for Macs with 8409 HDA chip + MAX98706/SSM3515 amps

https://github.com/davidjo/snd_hda_macbookpro

23 24 No sound 27"iMac Radeon RX 470/480 / 570/580/590] DigitalStereo (HDMI) / Newbie Corner / Arch Linux Forums

<https://bbs.archlinux.org/viewtopic.php?id=273816>

29 31 32 33 34 40 From Faceless to FaceTime: installing webcam drivers on a Debian-powered MacBook Air | Andrea Fortuna

<https://andreafortuna.org/2024/08/24/from-faceless-to-facetime-installing-webcam-drivers-on-a-debian-powered-macbook-air>

30 [Debian/Ubuntu] FacetimeHD Webcam Driver | GBAtemp.net

<https://gbatemp.net/threads/debian-ubuntu-facetimehd-webcam-driver.584652/>

35 mac - Ubuntu on iMac (2019 - 18,3) Crashes on Suspend - Ask Ubuntu

<https://askubuntu.com/questions/1224166/ubuntu-on-imac-2019-18-3-crashes-on-suspend>

36 linux-on-mac/mbpfan: A simple daemon to control fan ... - GitHub

<https://github.com/linux-on-mac/mbpfan>

37 Is macfanctld or mbpfan still needed for Mac fan support? - Reddit

https://www.reddit.com/r/linux_on_mac/comments/16tottk/is_macfanctld_or_mbpfan_still_needed_for_mac_fan/

38 39 Audio in 2019 iMac, Cirrus Logic CS8409 sound card [SOLVED]

<https://www.linuxquestions.org/questions/slackware-14/audio-in-2019-imac-cirrus-logic-cs8409-sound-card-%5Bsolved%5D-4175740788/>